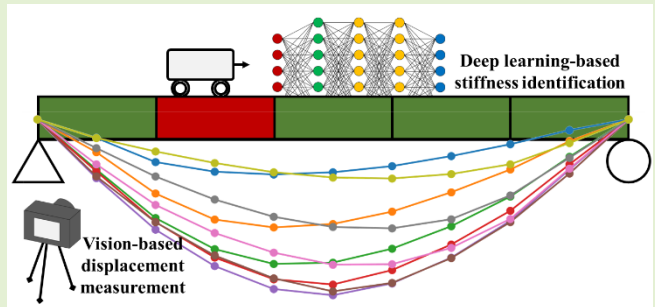


Finite Element Model Updating for Identifying Structural Stiffness in Bridge Structures via Computer Vision and Deep Learning

Yunwoo Lee^{ID}, Geonyeol Jeon, Namju Byun, and Hyungchul Yoon^{ID}

Abstract—Conventional structural maintenance approaches, such as visual inspections and vibration-based system identification, have limitations in consistency, reliability, and sensitivity to environmental factors. Simulation-based damage detection and dynamic response-based finite element (FE) model updating also have inherent constraints in practical applications. To address these challenges, this study proposes a novel methodology for identifying structural stiffness in bridge structures by integrating computer vision and deep learning techniques. The proposed method consists of two phases: 1) extracting displacement influence lines under a moving load using computer vision and 2) identifying stiffness distribution using deep learning. In the first phase, the computer vision technique accurately tracks vehicle positions and structural displacements, ensuring precise measurement of displacement influence lines. In the second phase, a deep neural network (DNN) updates the stiffness distribution of the FE model to reflect the as-is condition. The network is trained using simulation data with various stiffness distributions. A laboratory-scale test was conducted to validate the proposed method, demonstrating that the updated FE model closely matches the actual structure in terms of dynamic characteristics. The proposed method leverages influence lines to highlight the load–displacement relationship, which is directly linked to structural stiffness. By employing a well-trained DNN capable of handling various stiffness conditions, this approach enables the accurate estimation of key structural properties, facilitating effective data-driven structural health monitoring.

Index Terms—Computer vision, deep learning, finite element (FE) model updating, influence line, structural health monitoring (SHM).



I. INTRODUCTION

THERE are various methods to assess the condition of structures. The most commonly used methods are visual inspection and structural health monitoring (SHM). Visual inspection involves checking for visible damage or using simple testing equipment to detect damages in certain members [1], [2], [3]. This method provides a direct means to evaluate the current condition of the structure but can be

inconsistent and unreliable, as results may vary depending on the inspector. In addition, since the inspections are typically performed at intervals of several years, they have limitations in terms of continuous condition monitoring of the structure. SHM utilizes various sensors and data analysis techniques to estimate the condition of the structure and detect anomalies to prevent accidents [4], [5], [6]. This method can collect various responses, and the data can be analyzed through multiple approaches [7], [8], [9], [10], [11]. It offers more objective and comprehensive information about the structure, playing a crucial role in its condition management.

The most common method for evaluating the condition of a structure based on its response is the vibration-based system identification method using long-term measurement data [12], [13], [14], [15], [16]. This method analyzes the measured data in the frequency domain to extract dynamic characteristics such as the structural natural frequencies, which can be used to assess the condition of the structure. While this method is relatively simple to apply, it is limited by the fact that actual measurement data include the influence of various external

Received 31 March 2025; accepted 19 April 2025. Date of publication 30 April 2025; date of current version 16 June 2025. This work was supported by the National Research Foundation of Korea (NRF) funded by Korean Government (MSIT) under Grant RS-2022-NR071667 and Grant RS-2022-NR071137. The associate editor coordinating the review of this article and approving it for publication was Dr. Ke Feng. (Corresponding author: Hyungchul Yoon.)

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Digital Object Identifier 10.1109/JSEN.2025.3563927

factors, such as temperature, in addition to the structural behavior itself, making it difficult to analyze the structural characteristics independently of these external influences.

Damage detection is another method used to estimate changes in the structural condition based on its measured responses. This method is primarily developed through simulations with some experimental validation, where structural responses between the intact and damaged states are compared to estimate possible changes in the structural condition. To achieve this purpose, various structural responses, including acceleration, displacement, and strain, can be utilized [17], [18], [19], [20], [21], [22]. Unlike vibration-based system identification, which focuses on analyzing global dynamic characteristics of the structure, damage detection methods require advanced analytical techniques such as statistical analysis, signal processing, and artificial intelligence (AI) to clarify local damage in the structure [23], [24], [25], [26], [27], [28], [29], [30]. Although this technology is based on precise structural analysis simulations and experimental validation, various external factors, which are generally not considered in simulations or experiments, are included in the responses from real structures, thus limiting the applicability of the developed techniques. As a result, recent research has focused on unsupervised learning-based methods to overcome these challenges.

Finite element analysis (FEA) is one of the useful methods for analyzing the behavior of structures. However, even if the finite element (FE) model is constructed based on an actual structure, there will inevitably be differences between the theoretical model and the real structure. To overcome this limitation, the method of FE model updating has been introduced [31], [32], [33], [34], [35], aiming to develop a model that minimizes the discrepancy between the FEA-based behavior and the actual structural behavior. Typically, FE model updating is performed by extracting dynamic characteristics such as natural frequencies, damping ratios, and mode shapes from dynamic response data measured from the structure and then adjusting physical parameters such as stiffness and mass of the model through an iterative process to derive the optimal material properties. However, dynamic characteristics are inherently determined by the relationship between the stiffness and mass of the structure. In fact, when both the stiffness and mass of a structure increase at the same ratio, the structural resistance to external forces changes, but its dynamic characteristics remain unchanged. This is because dynamic characteristics, such as natural frequency, depend on the stiffness-to-mass ratio. Therefore, they may be useful for identifying the dynamic behavior of the structure, it is difficult to identify the material properties themselves, such as stiffness and mass, solely based on dynamic characteristics.

To more rationally determine the material properties of a structure, it is essential that factors directly related to these properties be available as known values. For instance, if the load acting on the structure can be used as a known value, the stiffness can be estimated from the resulting deflection. This is crucial information that dynamic responses alone cannot provide, and it can be used to improve the accuracy of the FE model. There are some studies that have utilized static response data from the structure to update the material proper-

ties of FE models [36], [37], [38]. However, they are primarily limited by their reliance on restricted measurement data, which prevents them from adequately considering responses under various loading conditions.

Several studies have utilized influence lines (or surfaces) to assess the effects of moving loads on structures [39], [40], [41], [42]. These studies have proposed more efficient methods for extracting influence lines or used them to estimate potential structural changes and assess the load-carrying capacity of bridges. One of the key advantages of influence lines is their ability to represent structural responses corresponding to different load positions, making them particularly useful for analyzing load-displacement relationships. This characteristic enables a more precise analysis of the structural system, facilitating accurate identification of stiffness distribution and overall structural behavior.

This study proposes a novel method for identifying the stiffness distribution of a structure by leveraging the advantages of influence lines in characterizing the quasi-static response of the structure. Unlike previous studies that have identified structural properties solely based on dynamic characteristics or have used influence lines primarily to evaluate moving loads, the proposed method utilizes influence lines as input data in an AI network to represent the quasi-static behavior of the structure for identifying its properties. Given that AI networks can effectively extract features from large datasets [43], [44], [45], [46], [47], an FEA simulation dataset generated under various structural conditions was used for training. Furthermore, computer vision technology was employed to simultaneously capture moving vehicle positions and structural displacements, enabling the precise extraction of displacement influence lines. This approach not only enhances the utilization of structural displacement data but also eliminates the need for time-consuming iterative optimization processes that often converge to multiple local solutions. Instead, a well-trained AI network, designed to account for various structural conditions, can instantly provide refined estimates of material properties based on precisely extracted influence lines that serve as structured input data.

The following section describes the methodology proposed in this study. Section III presents the experimental test conducted to validate the methodology, and Section IV discusses the results of FE model updating using the proposed method. Section V provides a brief conclusion summarizing the findings of this study.

II. METHODOLOGY

Fig. 1 provides a schematic description of the proposed methodology in this study. The method proceeds through three major stages. In the first stage, a moving vehicle load is applied to the target structure, and the displacement influence lines corresponding to the vehicle's position are extracted. To achieve this, a vision-based displacement measurement technique is utilized. Feature points for the vehicle and the locations on the structure, where displacement measurements are required, are identified and tracked, allowing for the measurement of both the vehicle's position and the structural displacement. Postprocessing identifies the frames when the vehicle reaches

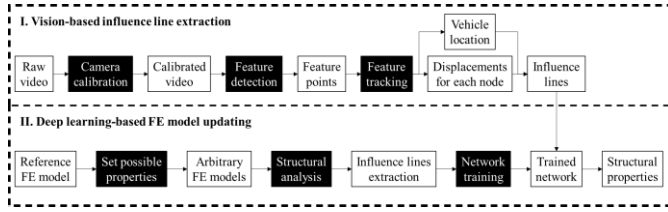


Fig. 1. Schematic description of the proposed method.

specific node points, from which the displacement influence lines for these key nodes are extracted.

In the second stage, considering potential stiffness degradation in the target structure, FE models with various stiffness distribution conditions are generated. FEA is then used to extract the displacement characteristics corresponding to the applied load positions. The displacements obtained from this analysis are postprocessed into the same format as the displacement influence lines from the first stage, making them suitable for training the artificial intelligence network.

Finally, in the third stage, the displacement influence lines extracted from the actual structure using the vision-based method are input into the trained network. The network, leveraging pretrained information, derives the stiffness distribution that best matches the actual deflection characteristics of the structure. This provides an updated FE model for analyzing the behavior of the target structure. In addition, the model itself can be used to estimate the stiffness distribution, indicating areas of stiffness degradation in the target structure.

A. Displacement Influence Lines

An influence line is defined as a graph that depicts the variations in responses, such as reactions at specific points within a structure, due to a single moving concentrated load traversing the entire structure. As shown in Fig. 2, when a unit load is applied to a simple beam, internal forces such as shear force and moment arise within the beam. Based on the equilibrium of forces, the reactions at supports A and B can be derived as expressed in (1). The moment $M(x)$ at position x , due to the load applied at position d , is given by (2). According to the Euler–Bernoulli beam theory [48], the relationship between the moment and the deflection w of the beam is expressed by (3). Using the appropriate boundary conditions, the deflection solution of the beam can be obtained through integration

$$\begin{cases} V_A = 1 - \frac{d}{L} \\ V_B = \frac{d}{L} \end{cases} \quad (1)$$

$$M(x) = \begin{cases} \left(1 - \frac{x}{L}\right)d, & \text{for } x > d \\ \left(1 - \frac{d}{L}\right)x, & \text{for } x \leq d \end{cases} \quad (2)$$

$$M(x) = -EI \frac{d^2 w}{dx^2} \quad (3)$$

where E represents the modulus of elasticity and I denotes the moment of inertia of the beam cross section. The displacement influence line derived in this manner provides significant

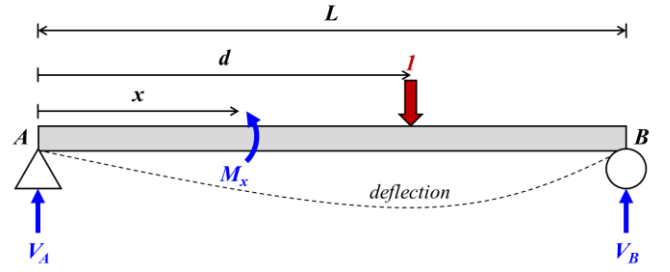


Fig. 2. Deflection and internal forces of a simply supported beam under unit load.

information about the structural response characteristics to the load positions, particularly for structures subjected to live or moving loads. Furthermore, since displacement due to external loads is directly linked to the stiffness of the structure, it serves as an informative factor for estimating structural stiffness in FE model updating.

Equation (3) is based on the assumption that the geometric and material properties are homogeneous throughout the beam, resulting in a constant flexural rigidity (EI). This assumption does not hold true for structures with heterogeneous characteristics or for those with localized damage. Therefore, this study utilizes FEA to account for such uncertainties. Influence lines were extracted based on models under various structural conditions, and these were used as training data for updating the FE model of the target structure.

B. Vision-Based Influence Line Extraction

In this study, the displacement influence line was extracted using a vision-based displacement measurement method. The procedure involved obtaining undistorted images from raw video, extracting feature points, and tracking them. Vehicle movement and beam deflection were measured based on the Kanade–Lucas–Tomasi (KLT) algorithm [49], [50], [51], and the influence line was derived through postprocessing.

1) *Camera Calibration*: The imaging device projects real-world objects onto an image plane, and the relationship between the actual 3-D coordinates and the image's 2-D coordinates is determined by the camera parameters. Equation (4) describes this relationship, where X , Y , and Z represent the world coordinates; x and y represent the image coordinates; and s is the scale factor. Parameters r_{ij} and t_i are the rotation matrix and translation vector, respectively, which describe the camera position. The parameters $f \cdot m_x$, c , and γ are intrinsic camera parameters representing the focal length, principal point, and skew coefficient, respectively,

$$\begin{aligned} s \begin{bmatrix} x & y & 1 \end{bmatrix} &= \begin{bmatrix} X & Y & Z & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \\ t_1 & t_2 & t_3 \end{bmatrix} \begin{bmatrix} f \cdot m_x & 0 & 0 \\ \gamma & f \cdot m_y & 0 \\ c_x & c_y & 1 \end{bmatrix}. \end{aligned} \quad (4)$$

Radial distortion is often applied to general imaging devices to increase the field of view. However, this distortion can

interfere with accurate displacement measurements. Therefore, for precise displacement measurements, it is essential to eliminate the effects of such distortion from the images. Camera calibration is the process designed to address this issue by extracting the camera parameters in (4). This allows for the correction of distortion effects in the raw video to produce rectified images. In this study, Zhang's [52] method was applied for camera calibration.

After obtaining the intrinsic parameters through camera calibration, these can be used to derive the radial distortion coefficients (k_1, k_2) for quantitatively analyzing the radial distortion in the image. Equation (5) represents the simplified relationship between the undistorted image and the distorted image

$$(x_d, y_d) = (x, y) \cdot (1 + k_1 r^2 + k_2 r^4) \quad (5)$$

where (x, y) denotes the undistorted coordinates, (x_d, y_d) denotes the distorted coordinates, and $r = (x^2 + y^2)^{1/2}$ represents the distance from the lens center. Using this radial distortion function, undistorted images can be obtained from raw video captured by the camera.

2) Feature Tracking: In the first frame of the calibrated video, the location of the structure where displacement is to be measured and the vehicle is set as a region of interest (ROI). Feature points within these ROIs are then extracted, and from the subsequent frames, the change of these feature points is used to measure the displacement at the target location. Various methods can be employed to extract feature points from ROIs; in this study, the Harris and Stephens [53] corner detector was used. The method determines whether a specific pixel is a corner point based on the intensity variations of surrounding pixels. It assumes that the transitional shift (x, y) is sufficiently small and defines the characteristics of the weighted sum of the squared intensity differences as described in the following equation:

$$\begin{aligned} E(\Delta x, \Delta y) &= \sum_{(x,y) \in w} [I(x + \Delta x, y + \Delta y) - I(x, y)]^2 \\ &= \sum_{(x,y) \in w} \left(\frac{\partial I}{\partial \Delta x} \Delta x + \frac{\partial I}{\partial \Delta y} \Delta y \right) \\ &= [\Delta x \quad \Delta y] M \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \end{aligned} \quad (6)$$

where Δx and Δy represent the movement of window W in x - and y -directions, respectively; I denotes the intensity; and M is defined as the structure tensor, as given in the following equation:

$$M = \sum_{(x,y) \in w} \begin{bmatrix} \left(\frac{\partial I}{\partial x} \right)^2 & \left(\frac{\partial I}{\partial x} \right) \left(\frac{\partial I}{\partial y} \right) \\ \left(\frac{\partial I}{\partial x} \right) \left(\frac{\partial I}{\partial y} \right) & \left(\frac{\partial I}{\partial y} \right)^2 \end{bmatrix}. \quad (7)$$

If λ_1 and λ_2 are the two eigenvalues of M , representing the directions of the maximum change in intensity gradients, a corner is detected only when both λ_1 and λ_2 have large positive values. The corner response R can be computed using the response function in (8). Here, k is an empirically determined constant, and $Tr(M)$ represents the trace of the

structure tensor M . The corner response R will exhibit high positive values in corner regions, negative values in edge regions, and small values in flat regions

$$R = \text{Det}(M) - k \text{Tr}(M)^2. \quad (8)$$

Once the target feature points are determined, changes in their positions in the image across frames can be tracked. Several tracking algorithms can be used for this purpose, and in this study, the KLT tracker was utilized. In the method, the relationship between the intensity $J(x)$ in the current frame and the intensity $I(x)$ in the previous frame can be expressed by (9), where d represents the displacement vector between two consecutive frames and g denotes the gradient vector

$$J(x) = I(x - d) = I(x) - g \cdot d \quad (9)$$

$$g = \left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right). \quad (10)$$

The residue ϵ for a window around the feature is defined by (11), where w denotes the weighting function. To minimize the residue, differentiating with respect to d yields (13), and the displacement vector d , which satisfies, is derived as shown in (14)

$$\begin{aligned} \epsilon &= \int [I(x - d) - J(x)]^2 w dA \\ &= \int [I(x) - g \cdot d - J(x)]^2 w dA = \int (h - g \cdot d)^2 w dA \end{aligned} \quad (11)$$

$$h = I(x) - J(x) \quad (12)$$

$$\frac{d\epsilon}{dd} = \int (h - g \cdot d) g w dA = 0 \quad (13)$$

$$d = \int g g^T w dA^{-1} \cdot \int (I - J) g w dA. \quad (14)$$

The KLT tracker tracks changes in the feature points within the defined ROI to determine the vehicle's position and the structural displacement at each position.

The response of the structure under the load of a moving vehicle was recorded on video. To define the region for extracting influence lines, the ends of the beam are set, and the video is segmented into intervals along the beam length. When the vehicle reaches a predefined region, the corresponding frame is designated as a keyframe, and the displacement is recorded. By performing this series of processes for all defined regions, the displacement influence line for the structure can be derived from the recorded displacements.

In this study, the vision-based displacement measurement method essentially followed Yoon et al. [54], and the Uni-KLT method [55] was employed to effectively measure the vertical deflection of the beam. Since the accuracy of the methods has been thoroughly validated in previous studies, further discussion on their accuracy is not addressed in this study.

C. Deep Learning-Based FE Model Updating

1) Training Data Acquisition Using FE Simulation: FE simulation targets structures used in the experimental test in Section III, aligning the positions of applied loads and the

TABLE I
REFERENCE VALUES OF MATERIAL PROPERTIES IN FE MODELS

Elastic modulus
205 Gpa

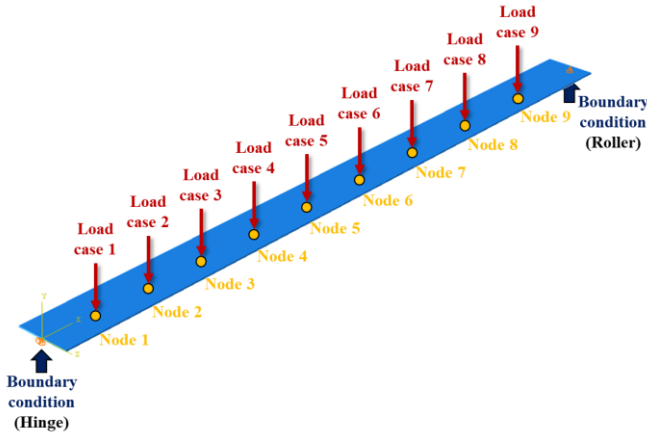


Fig. 3. FE model description.

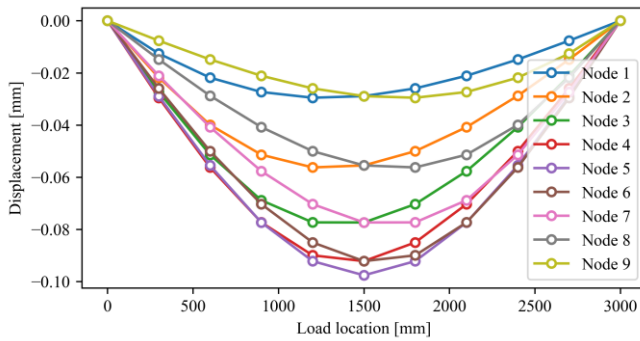


Fig. 4. Influence lines for the reference FE model.

displacement measurement nodes with those used in the experimental tests for extracting influence lines. The structural analysis program used is ABAQUS, and the model is constructed using 2-D beam elements. The geometric properties of the model were made to follow those in the experimental test, and the material properties were utilized from Table I as a reference. The boundary conditions applied are simply supported conditions. Fig. 3 illustrates the constructed FE model along with the applied loads and boundary conditions.

To extract displacements from each node, unit loads were applied at nine nodes identical to those used for displacement measurement. Influence lines for displacements on the constructed reference model are shown in Fig. 4.

The displacement data generated through structural analysis simulations are utilized as input data for training artificial intelligence networks. Here, the input data represent influence lines of displacements for nine nodes. The output data correspond to the stiffness distribution of the structure. The trained network is subsequently used for updating the properties of the FE model that align with the characteristics of actual structures based on extracted influence line information from target

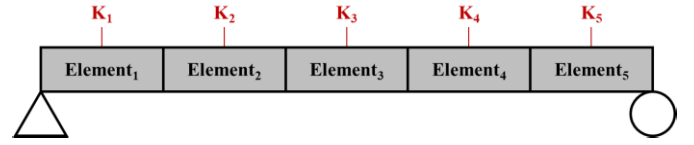


Fig. 5. Stiffness distribution in FE simulations.

TABLE II
APPLIED STIFFNESS OF FE MODELS IN THE SIMULATION TEST

Elastic modulus in FE simulation (K_i)		
Min. value	Reference value	Max. value
143.5 Gpa (70 %)	205 Gpa (100 %)	266.5 Gpa (130 %)

structures. Therefore, the simulation conditions for data generation should be rationalized based on engineering judgment. In this study, to enhance the features in the network training, the number of output data points was limited to be less than the number of input features. As shown in Fig. 5, the elements of the FE model for training the distribution of stiffness were divided into five parts. Each element was assigned a stiffness value ranging from 70% to 130% of the reference properties of the target structure.

Table II presents the reference values, maximum, and minimum values for the properties of the FE model used in the simulations. A total of 1000 simulation cases were considered, and in each case, influence lines of displacements were extracted at nine nodes for each load position.

2) *Network Training*: Artificial neural networks can be configured diversely depending on the objective. This study considers both the basic form of the artificial neural network, the deep neural network (DNN) model, and the long-short term memory (LSTM), which can learn sequential information from the data. In the DNN model, represented in Fig. 6(a), displacement data of nodes for each load case are simply distinguished and transmitted to the input layer, regardless of the node's position where load or displacement is extracted. In contrast, LSTM, a type of recurrent neural network (RNN), can learn the features of sequential data. In this study, the artificial intelligence network for updating the FE model is trained based on influence lines, which are displacement information of nodes according to the load positions acting on the target structure. These data represent structural response characteristics to sequential load positions. From this perspective, LSTM, capable of learning sequential data characteristics, was utilized, as shown in Fig. 6(b). The LSTM model processes displacement information according to load positions for each node in a distinct series format from the input layer, enhancing the informative features of the input data.

Table III represents the configuration of the artificial intelligence networks considered in this study regarding their input and output layers. In this study, the structure was divided into ten elements for the extraction of displacement influence lines. Since the nodes where loads are applied to the target structure and displacement extraction locations are nine in total, the LSTM model receives inputs in the form of $n \times 9 \times 9$. In contrast, the DNN model simply distinguishes and receives these

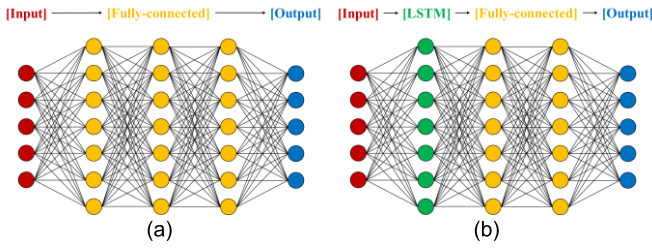


Fig. 6. Network models considered in this study. (a) DNN model. (b) LSTM model.

TABLE III
DATA CONFIGURATION IN INPUT-OUTPUT LAYERS

	Input layer (displacements)	Output layer (stiffness)
DNN model	$n \times 81$	
LSTM model	$n \times 9 \times 9$	$n \times 5$

TABLE IV
NETWORK TRAINING OPTIONS

Parameter	Value
Number of hidden layers	1 – 3
Number of hidden units	10 – 500
Total number of epochs	10,000
Early stop	1,000
Learning rate	0.001

inputs, resulting in $n \times 81$ inputs. The stiffness distribution is applied to the structure in five elements; thus, both models are designed to derive stiffness for five values in their output layers.

To analyze the training performance of the two networks considered in the study, the training options shown in Table IV were applied. Here, “the number of hidden layers” refers to the layers excluding the input and output layers. For the DNN model, this represents the number of stacked fully connected (FC) layers, while for the LSTM model, it includes both the LSTM and FC layers. Therefore, when the “number of hidden layers = 1,” the values passed through LSTM are directly transmitted to the output layer in the LSTM model. The “number of hidden units” was kept consistent for both LSTM and FC layers.

To ensure sufficient training, the total number of epochs was set to 50 000. For the efficient training process, an early stop option was applied with patience of 1000 epochs. This means that if the validation loss does not improve for 1000 consecutive epochs, the training process is terminated, even if there are more epochs left to run. The learning rate used for training was 0.001. Out of a total of 1000 simulation data, 80% were used for training, 10% for validation during training, and the remaining 10% for testing the models.

Fig. 7 demonstrates the training results for the network configuration variables considered. The DNN model showed that even with increasing numbers of layers and nodes, the validation loss did not improve significantly, remaining above 5×10^{-3} . This indicates that the network performance did not improve with the increase in model complexity.

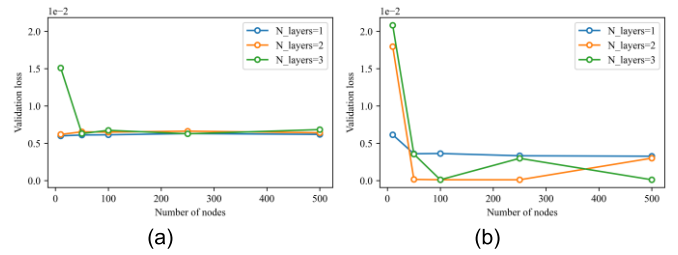


Fig. 7. Training performance for network parameters. (a) DNN model. (b) LSTM model.

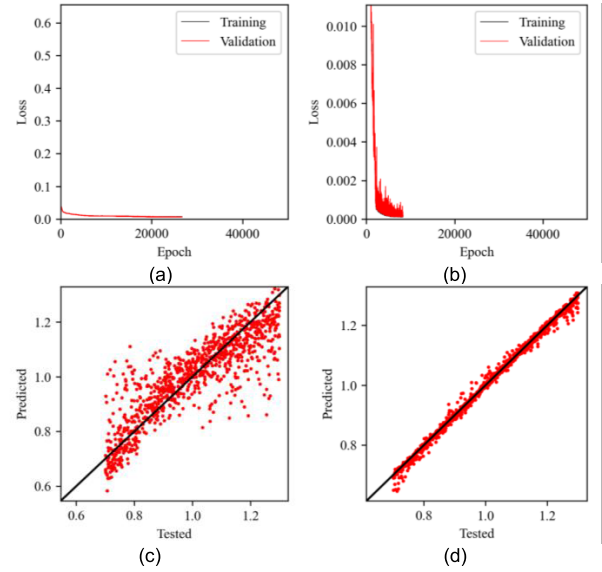


Fig. 8. Loss curves and prediction results of test data for the DNN and LSTM models. (a) Loss curve (DNN). (b) Loss curve (LSTM). (c) Prediction result (DNN). (d) Prediction result (LSTM).

In contrast, the LSTM model exhibited superior performance, achieving an MSE loss of approximately 1.11×10^{-4} when configured with two layers and 250 nodes. This result demonstrates that the LSTM model outperformed the DNN model in terms of the objective of this study.

Among the considered networks, the learning processes and test predictions of both models, which exhibited the best performance, are shown in Fig. 8. Both networks were sufficiently trained, with the DNN model reaching its best performance at the 24 689th epoch out of a total of 50 000 epochs, while the LSTM model showed its best performance at the 6285th epoch. In terms of prediction results on the test data, the DNN model exhibited an MSE loss of approximately 6.662×10^{-3} , while the LSTM model showed an MSE loss of about 1.287×10^{-4} . These results demonstrate that the DNN model was unable to achieve the intended performance in this study, while the LSTM model, which can more accurately reflect the sequential characteristics of the data, aligns with the objectives.

III. EXPERIMENTAL TEST

A. Experimental Setup

The proposed method was applied to the beam structure represented in Fig. 9. The model is 200 mm in width and

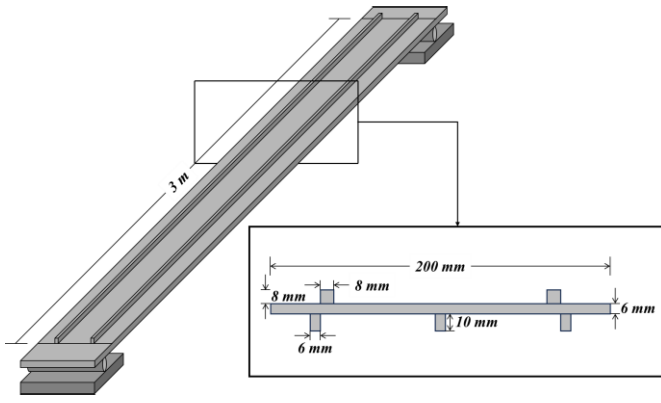


Fig. 9. Target model in experimental test.

TABLE V
SECTIONAL PROPERTIES OF THE TARGET
MODEL IN EXPERIMENTAL TEST

Area	Moment of inertia
1,508 mm ²	23,378 mm ⁴

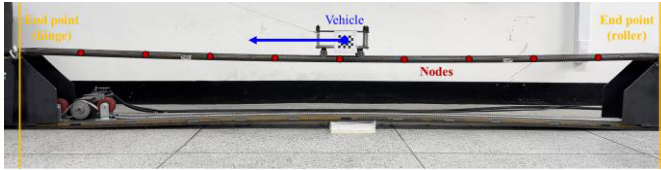


Fig. 10. Experimental setup.

has a total length of 3 m, spanning between the two support points. Two rails are attached to the top of the beam to accommodate vehicle movement and prevent derailing. One end of the beam is supported by a hinge, while the other end is simply placed on a support structure to simulate a simply supported condition.

Table V shows the structural specifications of the target model. Here, the cross-sectional area and the moment of inertia of the beam refer to the properties of the entire cross section, including the rails shown in Fig. 9.

Fig. 10 illustrates the overall experimental setup. A moving vehicle applies loads to the structure from the top, causing varying displacements along the structure. The displacements of the structure were measured at a total of nine node points. The positions of the vehicle and the resulting displacements of the structure were recorded using a camera installed at the front. The camera used in the experiment has a resolution of 3840×2160 and a frame rate of 30 frames/s.

B. Experiment Results

Fig. 11 shows the positions of vehicles extracted through computer vision. In this figure, the x -direction position represents the manually controlled vehicle position, while the y -direction position indicates changes in vertical position due to beam deflection. The results demonstrate differences in the y -direction position between the start and end points of vehicle movement. Possible causes include height differences at both ends of the beam in the experimental setup, the image itself not

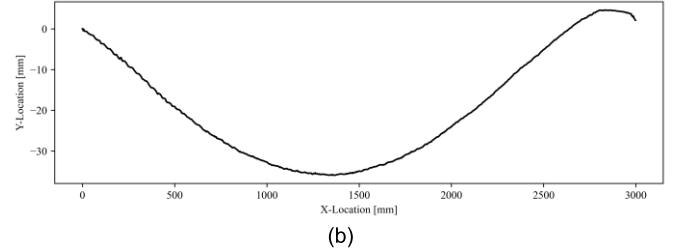
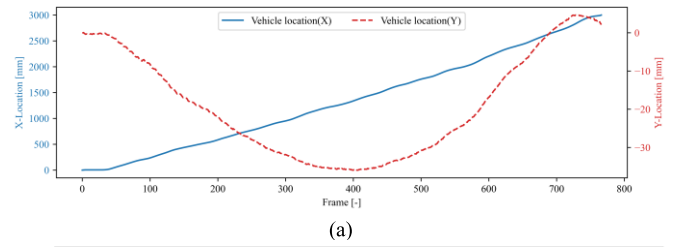


Fig. 11. Vehicle positions extracted using the vision-based displacement measurement method. (a) Vehicle position x and y per frame. (b) Position y of the vehicle as a function of position x .

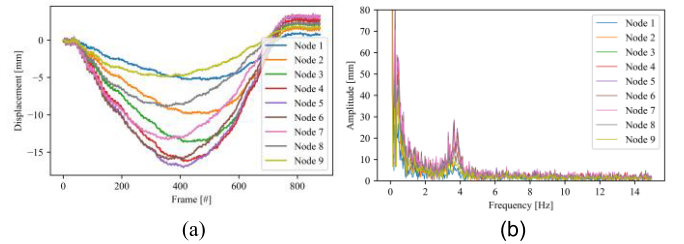


Fig. 12. Displacements and frequency-domain characteristics of each node. (a) Displacements measured at each node. (b) Fast Fourier transform of measured displacements.

being perfectly horizontal with the beam, or cumulative errors during tracking due to the nature of the KLT tracker method. The sudden drop in the vehicle's y -direction position near the end is attributed to the loss of feature points for tracking as the vehicle moves out of the image frame.

To extract the influence lines of displacements due to vehicle movement, the beam was divided into ten elements, and displacements were measured at nine nodes excluding the end points. The displacements at each node are shown in Fig. 12(a). The displacements indicate variations in beam deflection due to vehicle movement, coupled with inherent dynamic components of the beam itself.

To analyze the frequency-domain characteristics of the extracted displacements, a fast Fourier transform was applied, and the results are presented in Fig. 12(b). Near 0 Hz in the spectrum, the low-frequency components reflect the deflection characteristics of the beam caused by the slow movement of the vehicle. The peak frequency components between 2.5 and 5.0 Hz correspond to the natural frequencies of the vibration modes.

The displacement influence line represents the variation in static displacements with respect to the position of the load. Therefore, the dynamic components included must be excluded from these displacements. Consequently, a low-pass filter was applied to remove dynamic components from each displacement. The filtered displacements, along with their

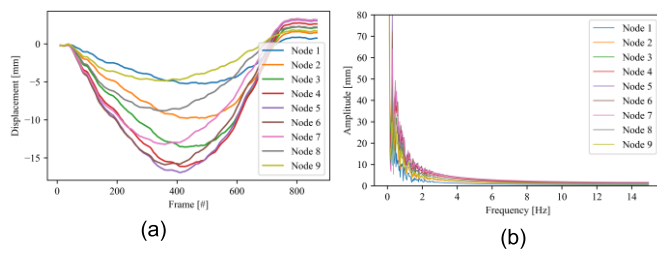


Fig. 13. Filtered displacements and frequency-domain characteristics. (a) Filtered displacements at each node. (b) Fast Fourier transform of filtered displacements.

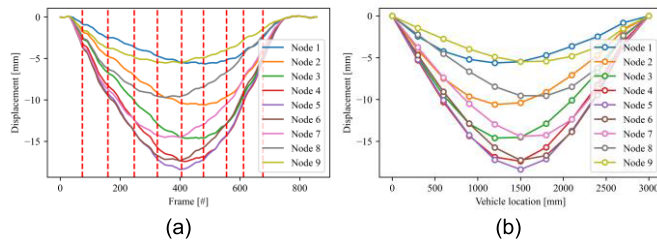


Fig. 14. Detrended node displacements and extracted displacement influence lines. (a) Detrended displacements at each node. (b) Displacement influence lines extracted from experiments.

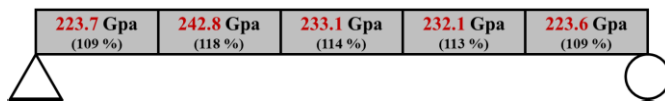


Fig. 15. Updated stiffness distribution.

corresponding frequency-domain characteristics, are shown in Fig. 13. While these displacements appear relatively stable compared to before the removal of dynamic components, they still contain trend components significant enough to hinder analysis in their original form. This discrepancy manifests as inconsistencies in displacements at the start and end points of the frame. To mitigate these effects, detrending was applied to the data.

Detrended node displacements are shown in Fig. 14(a). The red dashed lines in the figure represent the frames corresponding to the moments when the vehicle reaches each node point, from which displacements are extracted to derive the influence lines. The displacement influence lines of the target structure extracted through experiments are shown in Fig. 14(b).

IV. FE MODEL UPDATING

The stiffness distribution of the target structure was derived by applying the displacement influence lines extracted from the experimental test to the network trained through FE simulation data. The estimated stiffness of the structure appears to range approximately between 223.7 and 242.8 GPa (109%–118%) as shown in Fig. 15, which is slightly higher than the reference value of 205 GPa (100%) used in the FE model.

To examine the dynamic characteristics of the updated FE model, natural frequencies were analyzed as presented in Table VI. The stiffness of the updated model was determined using the proposed method. The mass was directly measured and applied as 34.8 kg, which represents a decrease

Mode shape	Natural Frequency [Hz]	
	Reference	Updated
1 st mode	3.517	3.797
2 nd mode	14.128	15.192
3 rd mode	32.026	34.119

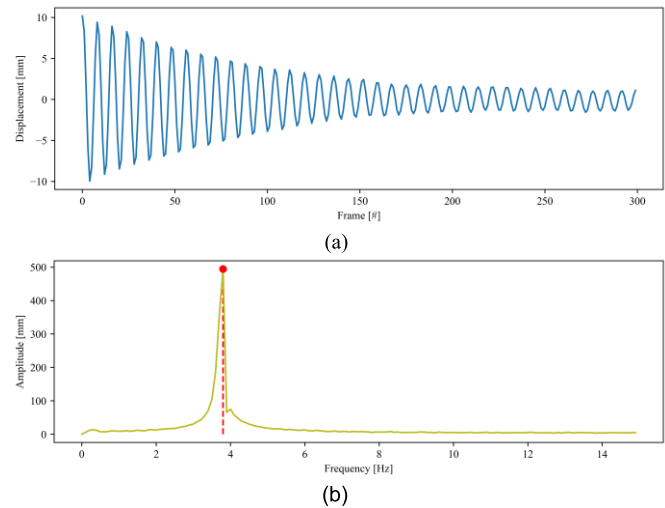


Fig. 16. Dynamic characteristics of target model in experimental test. (a) Free vibration displacement. (b) FFT of free vibration displacement.

of approximately 2% compared to the reference value of 35.51 kg. To compare before and after model updating, the natural frequency was compared with the model to which the reference material properties were applied. The updated FE model showed an increase of about 7% in the first, second, and third eigenfrequencies compared to the reference model before updating.

To assess whether the updated FE model reasonably represents the dynamic characteristics of the target structure, a free vibration test was conducted. Fig. 16 presents the free vibration displacements measured through experimental tests on the target structure, along with frequency-domain analysis results. The frame rate of the camera used in this study was 30 Hz; therefore, frequencies above half of the sampling rate

(i.e., above 15 Hz, according to Nyquist theory) were not analyzed.

From the free vibration test, the first natural frequency of the target structure was extracted and measured at approximately 3.8 Hz. This value shows a slight difference of about 0.08% from the eigenfrequency of 3.797 Hz obtained from the updated FE model. The structural response used in this study corresponds to deflection variations in the structure due to vehicle movements, representing quasi-static behavior, excluding dynamic characteristics. Although the dynamic characteristics were not explicitly considered in the FE model update, the updated model demonstrated results closely aligned with the actual dynamic characteristics of the structure.

V. CONCLUSION

In this study, a vision-based FE model updating method was proposed using displacement influence lines for moving loads with artificial intelligence technology. This approach involves training artificial intelligence networks on simulation data for various structural conditions and deriving properties of FE models corresponding to the target structure based on measured load positions and structural displacement information through computer vision. The applicability of the proposed method was validated through experimental tests, leading to the following conclusions.

- 1) Using vision-based tracking methods, the positions of moving loads and the displacements of structures can be measured, and postprocessing such as filtering can be applied to ensure data integrity.
- 2) The DNN model, which is the basic artificial neural network, showed limited performance in deriving the properties of structures based on displacement information. To address this, an LSTM-based model was employed to extract features of sequential data, such as influence lines. It showed compatible performance for the purpose of the proposed method.
- 3) The displacement influence lines utilized in the proposed method represent quasi-static behavior, excluding dynamic characteristics of the structure. Nevertheless, the derived FE model demonstrated a difference of just 0.08% in the first natural frequency of the target structure, proving the effectiveness of the proposed method.

The proposed method utilizes the external loads applied to the structure as known values and based on the corresponding response enables the estimation of stiffness distribution, which is one of the key structural properties. This cannot be achieved solely through the dynamic response as in conventional methods, and it can significantly contribute to the advancement of rational FE model updating techniques. Moreover, recent advancements in machine learning and signal processing techniques, especially in industrial applications, provide new opportunities to enhance the accuracy and versatility of FE model updating methodologies.

While the proposed method has been experimentally validated, several practical challenges remain for real-world applications. Capturing the entire structure for vision-based measurement and obtaining influence lines for multispan bridges are key challenges that need to be addressed. Future

research should focus on developing technologies to overcome these practical limitations. In addition, further studies are needed to develop more accurate vision-based displacement measurement methods and to update structural properties, such as mass and geometric characteristics, which were not considered in this study, in order to achieve a more accurate and advanced FE model updating framework.

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